This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): TH671ST

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
3.04.0001-b002	2.3.1

Version History

Module Version	Date	Notes
1_0_0_0	8/22/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
`InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='TH671ST')`
SerialOverEthernetClass
`InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='TH671ST')`

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
3DMode	'Off' 'Frame Sequential' 'Inverter Disable'	'Auto' 'Frame Packing' 'Inverter Enable'	'Top Bottom' 'Side by Side'
`# 3DMode example` `InterfaceName.Set('3DMode', 'Off')`			
AspectRatio	'Auto' 'Wide'	'4:3' 'Letterbox'	'16:9'
`# AspectRatio example` `InterfaceName.Set('AspectRatio', 'Auto')`			
AudioMute	'On'	'Off'	
`# AudioMute example` `InterfaceName.Set('AudioMute', 'On')`			
AutoImage [1]	None		
`# AutoImage example` `InterfaceName.Set('AutoImage', None)`			
Freeze	'On'	'Off'	
`# Freeze example` `InterfaceName.Set('Freeze', 'On')`			
Input	'Computer'	'HDMI 1'	'HDMI 2'
`# Input example` `InterfaceName.Set('Input', 'Computer')`			
LampMode	'Normal'	'Eco'	'Smart Eco'
`# LampMode example` `InterfaceName.Set('LampMode', 'Normal')`			
MenuCall	'On'	'Off'	
`# MenuCall example` `InterfaceName.Set('MenuCall', 'On')`			
MenuNavigation	'Up' 'Right'	'Down' 'Enter'	'Left'
`# MenuNavigation example` `InterfaceName.Set('MenuNavigation', 'Up')`			
PictureMode [2]	'Bright' 'Game'	'Vivid' 'User 1'	'Cinema' 'User 2'
`# PictureMode example` `InterfaceName.Set('PictureMode', 'Bright')`			

Command	Value	Value
Power	'On'	'Off'

```
# Power example
InterfaceName.Set('Power', 'On')
```

Command	Value	Value
VideoMute	'On'	'Off'

```
# VideoMute example
InterfaceName.Set('VideoMute', 'On')
```

Command	Value	Value
VolumeStep	'Up'	'Down'

```
# VolumeStep example
InterfaceName.Set('VolumeStep', 'Up')
```

[1] Available when the Input is Computer.

[2] Available when 3D Mode is Off.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
3DMode	'Off' 'Frame Sequential' 'Inverter Disable'	'Auto' 'Frame Packing' 'Inverter Enable'	'Top Bottom' 'Side by Side'

```
# 3DMode example
InterfaceName.Update('3DMode')
Value = InterfaceName.ReadStatus('3DMode')
InterfaceName.SubscribeStatus('3DMode', None, FeedbackHandler)
```

Command	Value	Value	Value
AspectRatio	'Auto' 'Wide'	'4:3' 'Letterbox'	'16:9'

```
# AspectRatio example
InterfaceName.Update('AspectRatio')
Value = InterfaceName.ReadStatus('AspectRatio')
InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)
```

Command	Value	Value
AudioMute	'On'	'Off'

```
# AudioMute example
InterfaceName.Update('AudioMute')
Value = InterfaceName.ReadStatus('AudioMute')
InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)
```

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
# ConnectionStatus example
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value
Freeze	'On'	'Off'

```
# Freeze example
InterfaceName.Update('Freeze')
Value = InterfaceName.ReadStatus('Freeze')
InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)
```

Command	Value	Value	Value
Input	'Computer'	'HDMI 1'	'HDMI 2'

```
# Input example
InterfaceName.Update('Input')
Value = InterfaceName.ReadStatus('Input')
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value	Value
LampMode	'Normal'	'Eco'	'Smart Eco'

```
# LampMode example
InterfaceName.Update('LampMode')
Value = InterfaceName.ReadStatus('LampMode')
InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)
```

Command	Value
LampUsage	Hours

```
# LampUsage example
InterfaceName.Update('LampUsage')
Value = InterfaceName.ReadStatus('LampUsage')
InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)
```

Command	Value	Value	Value
PictureMode [1]	'Bright'	'Vivid'	'Cinema'
	'Game'	'User 1'	'User 2'

```
# PictureMode example
InterfaceName.Update('PictureMode')
Value = InterfaceName.ReadStatus('PictureMode')
InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)
```

Command	Value	Value
Power	'On'	'Off'

```
# Power example
InterfaceName.Update('Power')
Value = InterfaceName.ReadStatus('Power')
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Command	Value	Value
VideoMute	'On'	'Off'

```
# VideoMute example
InterfaceName.Update('VideoMute')
Value = InterfaceName.ReadStatus('VideoMute')
InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)
```

Command	Value
VolumeStatus	0 – 100

```
# VolumeStatus example
InterfaceName.Update('VolumeStatus')
Value = InterfaceName.ReadStatus('VolumeStatus')
InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)
```

[1] Available when 3D Mode is Off.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

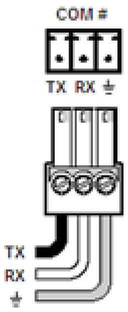
Notes for the Device

To avoid entering a password each time the projector powers on, set Advanced Menu - System Setup: Advanced → Password → **Power On Lock** to **Off**.

Serial communication

Port Type:	RS-232	**Parity:**	None
Baud Rate:	115200	**Stop Bits:**	One
Data Bits:	8	**Flow Control:**	None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	⟶	2	RxD
RxD	⟵	3	TxD
GND	——	5	GND

Appendix A. Set Commands

3D Mode Auto	\x0D*3d=auto#\x0D
3D Mode Frame Packing	\x0D*3d=fp#\x0D
3D Mode Frame Sequential	\x0D*3d=fs#\x0D
3D Mode Inverter Disable	\x0D*3d=da#\x0D
3D Mode Inverter Enable	\x0D*3d=iv#\x0D
3D Mode Off	\x0D*3d=off#\x0D
3D Mode Side by Side	\x0D*3d=sbs#\x0D
3D Mode Top Bottom	\x0D*3d=tb#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Letterbox	\x0D*asp=LBOX#\x0D
Aspect Ratio Wide	\x0D*asp=WIDE#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input Computer	\x0D*sour=RGB#\x0D
Input HDMI 1	\x0D*sour=hdmi#\x0D
Input HDMI 2	\x0D*sour=hdmi2#\x0D
Lamp Mode Eco	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Lamp Mode Smart Eco	\x0D*lampm=seco#\x0D
Menu Call Off	\x0D*menu=off#\x0D
Menu Call On	\x0D*menu=on#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
Picture Mode Bright	\x0D*appmod=bright#\x0D
Picture Mode Cinema	\x0D*appmod=cine#\x0D
Picture Mode Game	\x0D*appmod=game#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D
Picture Mode Vivid	\x0D*appmod=vivid#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D
Volume Step Down	\x0D*vol=-#\x0D

Volume Step Up	`\x0D*vol=+#\x0D`

Appendix B. Update Commands

3D Mode	`\x0D*3d=?#\x0D`
Aspect Ratio	`\x0D*asp=?#\x0D`
Audio Mute	`\x0D*mute=?#\x0D`
Freeze	`\x0D*freeze=?#\x0D`
Input	`\x0D*sour=?#\x0D`
Lamp Mode	`\x0D*lampm=?#\x0D`
Lamp Usage	`\x0D*ltim=?#\x0D`
Picture Mode	`\x0D*appmod=?#\x0D`
Power	`\x0D*pow=?#\x0D`
Video Mute	`\x0D*blank=?#\x0D`
Volume Status	`\x0D*vol=?#\x0D`